Skills Summary

Right-Triangle Trigonometry — Choosing and Checking

Use this reference while completing your worksheet or when studying.

Skill 1 — Pick the ratio that fits your two sides

SOH-CAH-TOA: $\sin A = \text{opp}/\text{hyp}$, $\cos A = \text{adj}/\text{hyp}$, $\tan A = \text{opp}/\text{adj}$.

Method: label the sides *from the angle you are working from* — opposite, adjacent, hypotenuse — then pick the ratio that contains the side you have and the side you want. The hypotenuse is always opposite the right angle; it is never the adjacent side.

Special triangles: 30-60-90 has sides $1 : \sqrt{3} : 2$ (shortest opposite the 30°); 45-45-90 has $1 : 1 : \sqrt{2}$.

Example: The hypotenuse is 15 and the angle is 28°. Find the side opposite that angle.
Step 1. Want the opposite side, know the hypotenuse — opposite with hypotenuse is sine, the S of SOH.
Step 2. $\sin 28° = x/15$, so $x = 15 \sin 28° = 7.0420\ldots$ with the calculator in degree mode. $\boldsymbol{x \approx 7.04}$, and it is shorter than the hypotenuse, as a leg must be.

Try it: The hypotenuse is 18 and the angle is 40°. Find the leg adjacent to that angle.

check: **13.79**

Skill 2 — Inverse trig for a missing angle

Two sides given, angle wanted: build the ratio first, then apply the inverse: $A = \sin^{-1}(\text{opp}/\text{hyp})$, $A = \cos^{-1}(\text{adj}/\text{hyp})$, $A = \tan^{-1}(\text{opp}/\text{adj})$.

Two different things: $\sin^{-1}$ is the INVERSE (it returns an angle); $(\sin A)^{-1}$ is the RECIPROCAL, $1/\sin A$ (it returns a number). The calculator button marked $\sin^{-1}$ is the inverse.

Upside-down check: the bigger side always faces the bigger angle. If the side opposite your angle is the shorter one, your answer must be under 45°.

Example: The side opposite A is 5 and the hypotenuse is 13. Find $\angle A$.
Step 1. Two sides are known and an angle is wanted, and the pair is opposite-with-hypotenuse, so it is the inverse sine that undoes it.
Step 2. $\sin A = 5/13$, so $A = \sin^{-1}(5/13) = 22.6198\ldots$ degrees. $\boldsymbol{A \approx 22.62°}$, under 45° as the short opposite side demands.

Try it: The side adjacent to A is 8 and the hypotenuse is 17. Find $\angle A$.

check: **61.93°**

Skill 3 — Degree mode and the sanity check

Set the mode first. Every angle on this sheet is in degrees, so the screen must read DEG. In radian mode the calculator answers a different question and gives no warning.

Two checks that catch a mode error instantly: $\sin 30°$ must read exactly 0.5, and a leg must come out shorter than the hypotenuse. A negative length, or a length larger than the hypotenuse, is a mode error until proven otherwise.

Example: The leg adjacent to a 25° angle is 30. Find the opposite leg.
Step 1. Opposite over adjacent is tangent, so the setup is a tangent — and tangent is where a mode error does the most damage, so check DEG before pressing enter.
Step 2. $30 \tan 25° = 13.9892\ldots$ in degree mode; the same keystrokes in radian mode return -4.01, which no length can be. **13.99**

Try it: The leg adjacent to a 63° angle is 12. Find the opposite leg.

check: **23.55**

(!) Watch out: a wrong ratio usually returns a believable number — sine and cosine of the same angle both give a length shorter than the hypotenuse. Size checks catch mode errors; only naming opposite and adjacent catches a wrong ratio.